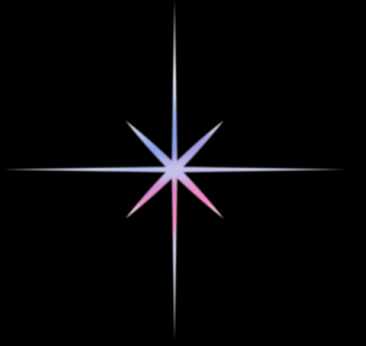


# HUMAN ACTIVITY RECOGNISATION

IOT\_Project'26





# Project overview & objectives

## What the project is about ?

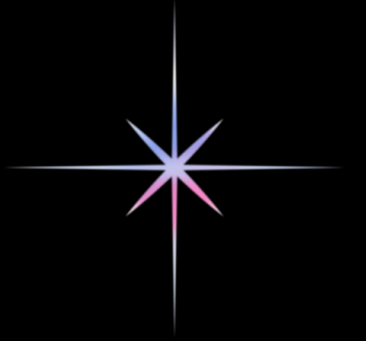
our project focuses on the Human activity recognition (HAR) through which we intend to identify few physical states like :

- 1.Sitting
- 2.Standing
- 3.walking
- 4.Running

## Our Objective ?

our goal is to calculate a final activity score to show how active the person is during a monitored time stamp





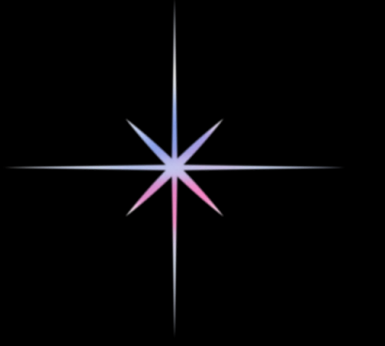
## Why this project ?

While the existing fitness trackers and apps are well designed and are delivering great data analysis , but most of them are not beginner friendly and for the people who are into tech jobs would mostly sit and work without doing much intense workouts & they are usually little new to such intense workouts.

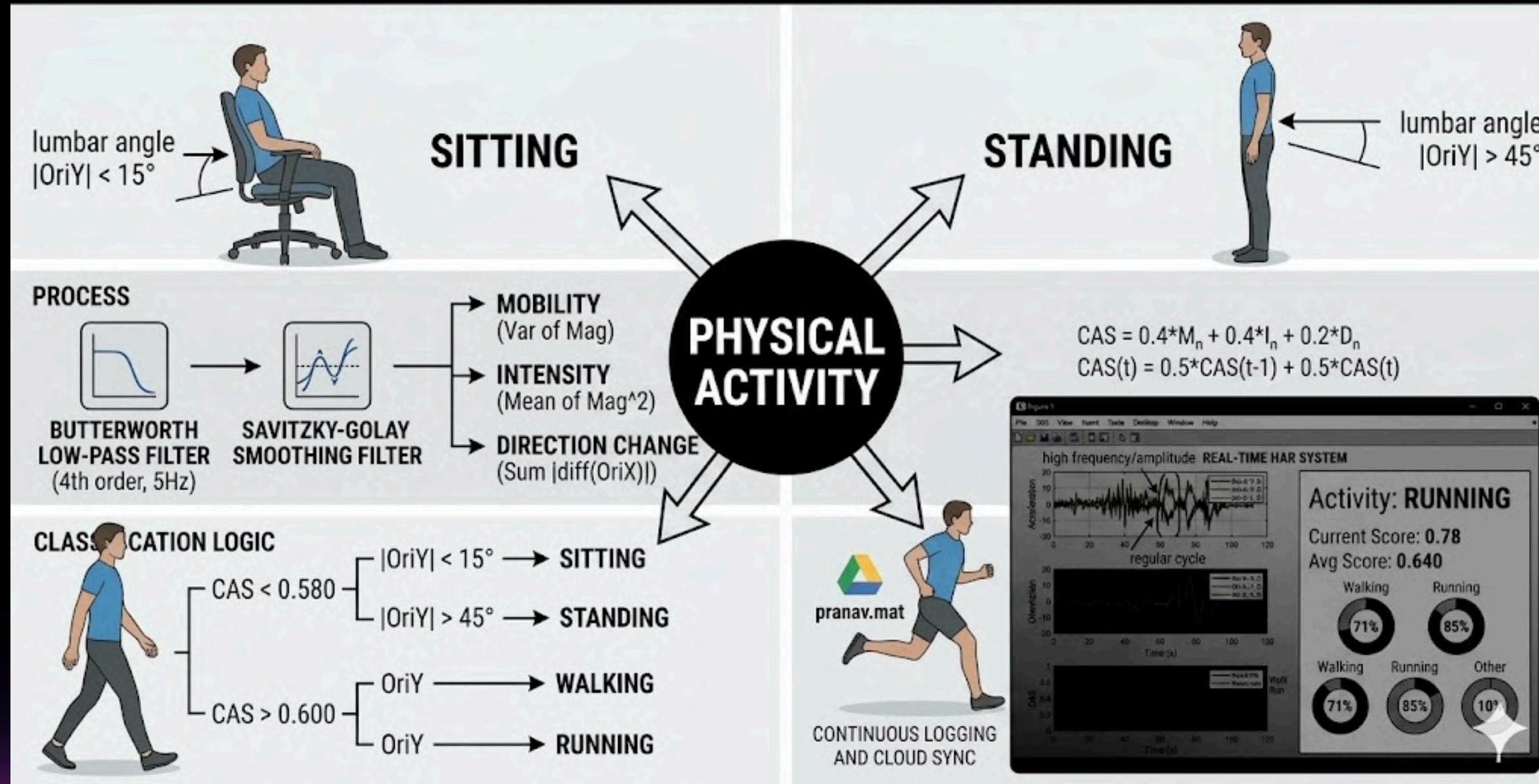
So to address this we came up with an solution ;through which people can analyse their activity score and can work more on their fitness goals , unlike the traditional deliverables we here intend to give them an activity estimate of them doing few basic physical works like walking , standing or Running and finally they can also understand their maximum percentage of jobs performed on which of these activities.



# Physical Activity



## REAL-TIME HAR CLASSIFICATION FROM SENSOR DATA

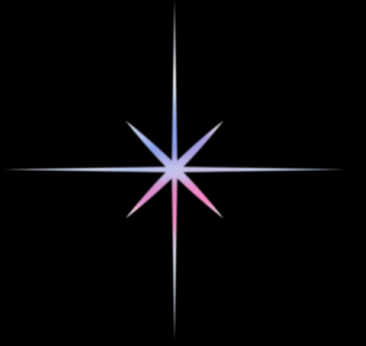




HUMAN ACTIVITY



# Mathematical Parameters



DATA COLLECTION

we've used the Matlab mobile sensors specifically the acceleration & orientation sensors

Co-ordinate Independence

Because the mobile orientation can be changed (one of the possibilities is due to its presence in pockets ) we calculated the magnitude of acceleration across all 3 axes.

$$(Mag = \sqrt{X^2 + Y^2 + Z^2})$$

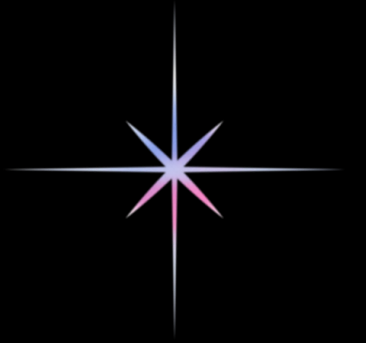
Dual Sensors

We synchronised data from both the sensors , orientation ( Specifically the Y- tilt ) which acts as posture detector between sitting & standing , while acceleration acts as motion detector.

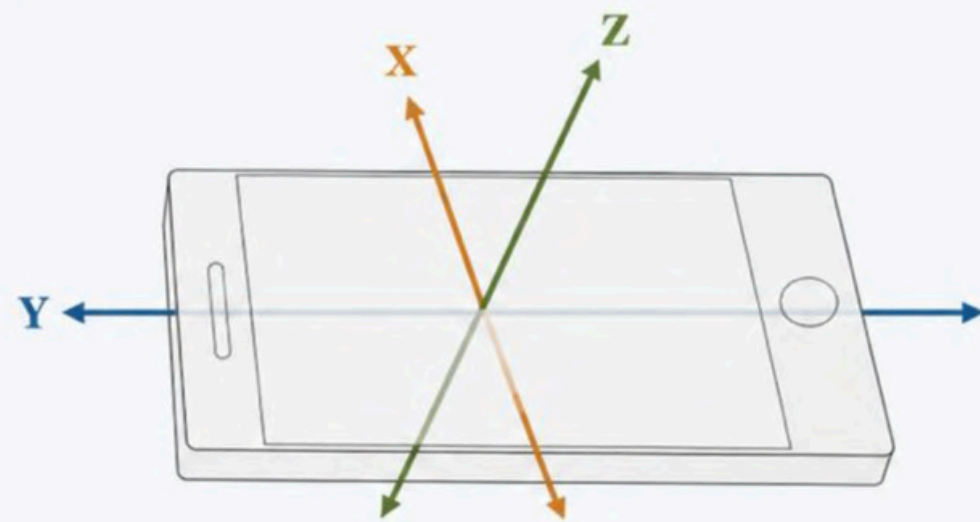


# Sensors

## ACCELERATION SENSOR



### ACCELEROMETER SENSOR



Measures linear acceleration along the three axes.

**X-axis (orange):** Left ↔ Right

**Y-axis (blue):** Front ↔ Back

**Z-axis (green):** Up ↔ Down

**Outputs (includes gravity + motion):**

- $A_x$  : Acceleration along X-axis ( $\text{m/s}^2$ )
- $A_y$  : Acceleration along Y-axis ( $\text{m/s}^2$ )
- $A_z$  : Acceleration along Z-axis ( $\text{m/s}^2$ )

#### Example Readings

$A_x = +0.35 \text{ m/s}^2$   
(moving right)

$A_y = -0.12 \text{ m/s}^2$   
(moving back)

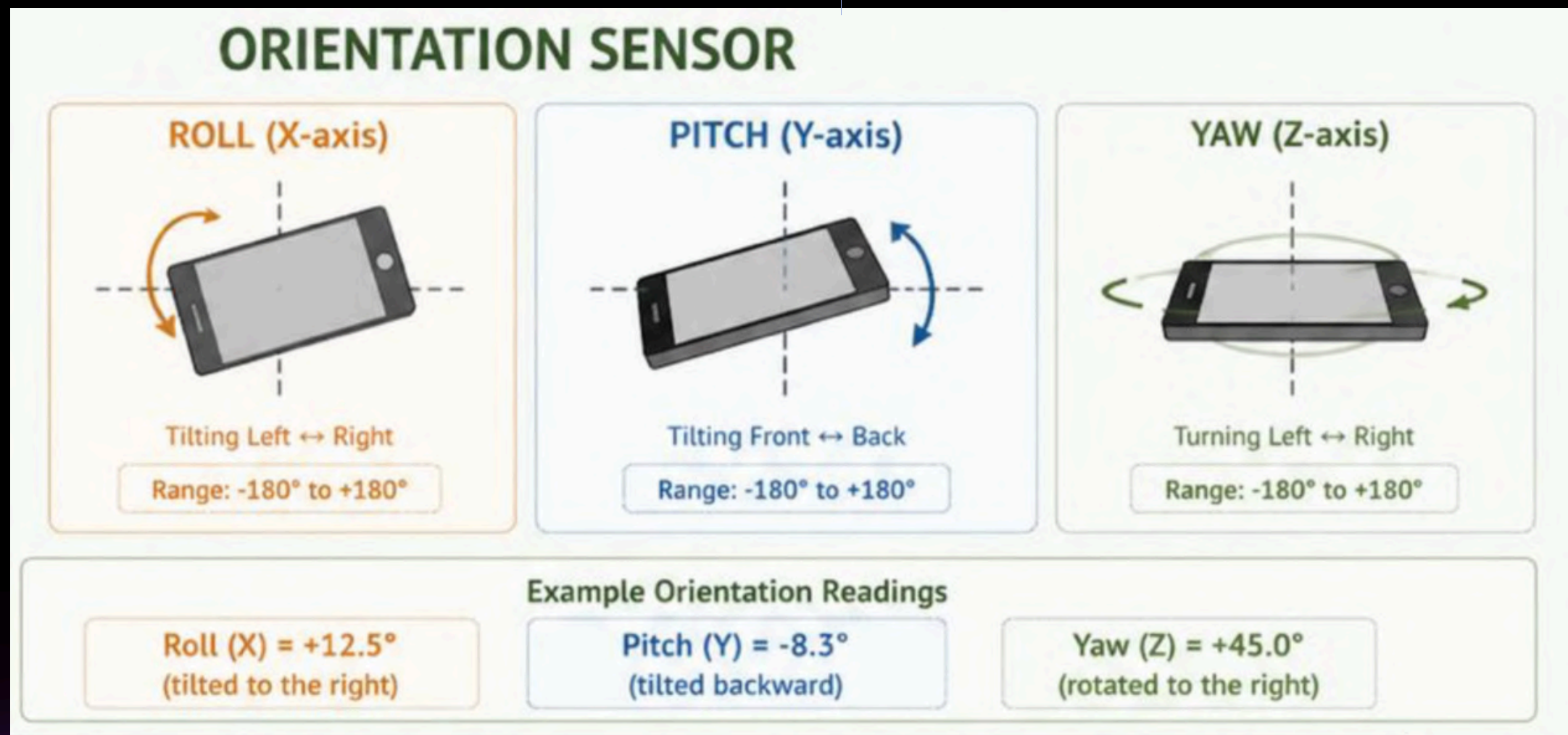
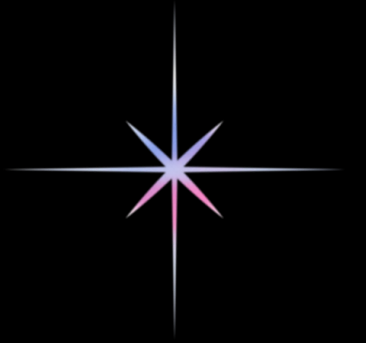
$A_z = +9.78 \text{ m/s}^2$   
(phone flat, facing up)

We've measured the acceleration across all 3 axis while the X & Y axis being the variable acceleration and Z being constant acceleration . and finally the magnitude of acceleration is being used in both intensity and mobility parameters



# Sensors

## ORIENTATION SENSOR



Measures the device orientation majorly in the 2 axis X-axis and Y-axis while the X axis being the major stakeholder in direction change ; and for the activity classification Y-aixes is being considered





## ANALYSIS

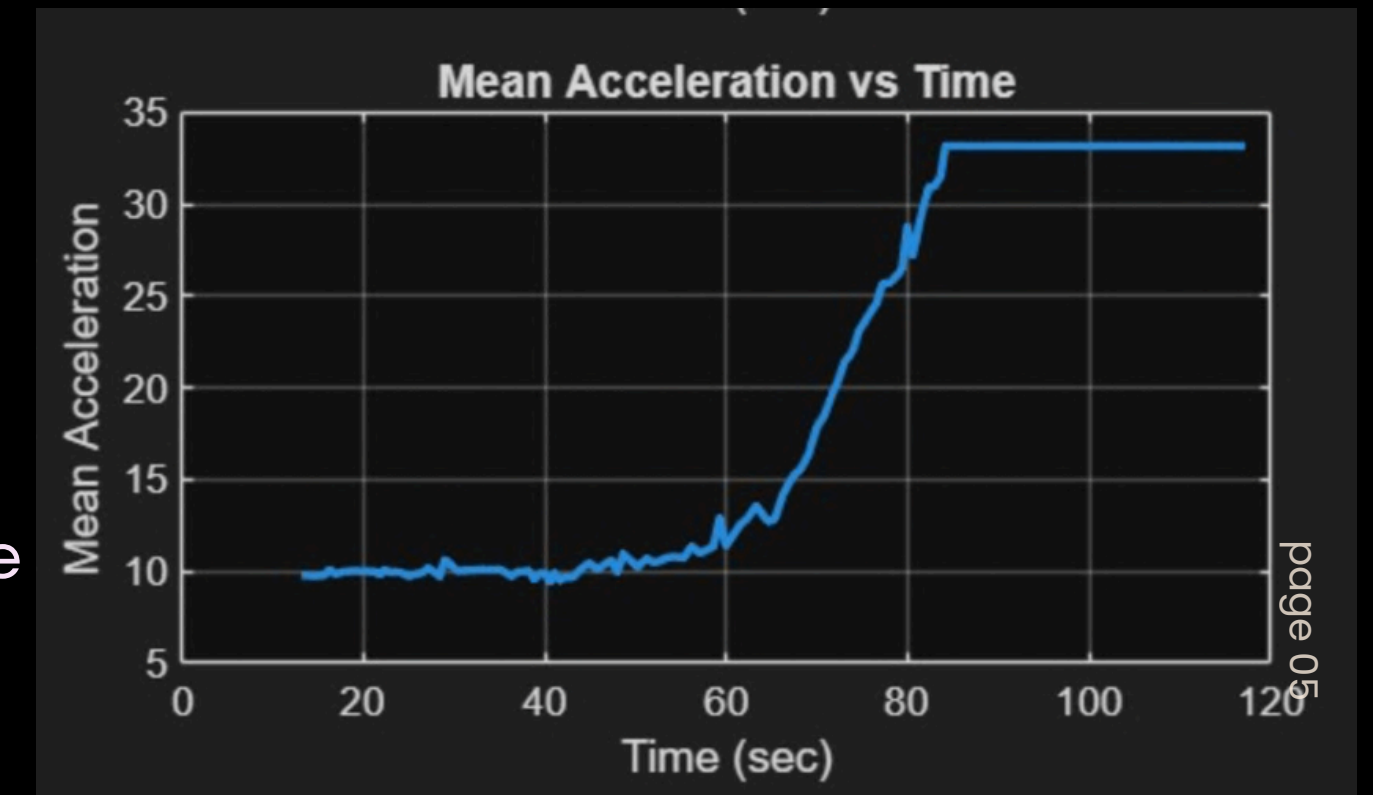
We initially tried & explored calculating jerk ( the derivative of acceleration ) and step frequency (dividing the number of detected signal peaks by the window duration ) , Variance , mean & RMS & finally we've realised that we can solve the cases without jerk and step frequency ( because these 2 parameters increase the complexity and rely only on acceleration peaks )

### Mathematics involved

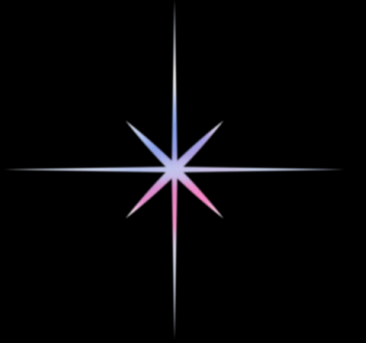
**Mobility:** we calculated this using variance of the filtered acceleration through this we can measure the fluctuations of tasks and this variance parameter would help in identifying the task switch and also the tasks intensity variation

**Intensity:** we measured this using the mean square of the signal to measure the intensity of tasks

**Direction change:** Through the absolute difference in the X-axis orientation we measured the user turning

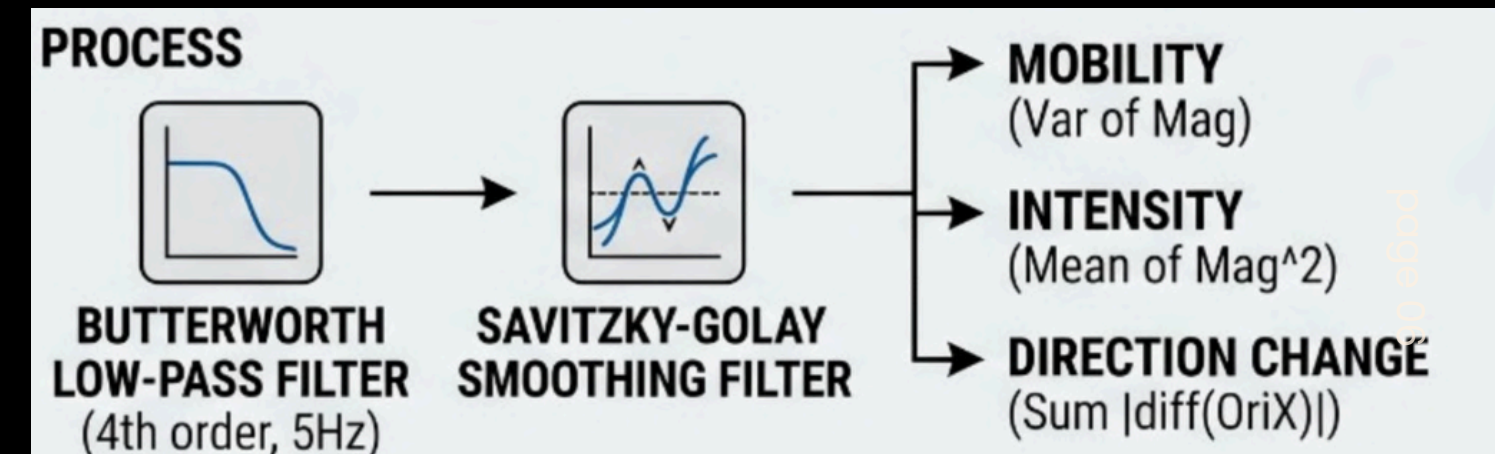


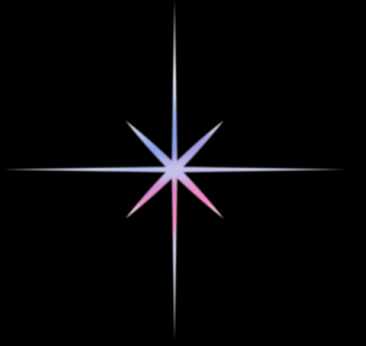




# Signal processing & filters :

- **Removing Noise** : we first used an 4<sup>th</sup> order butterworth low pass filter with an 5Hz cut-off ( because most of significant human activity occurs below 5Hz) & this 5Hz cutoff would help us clear most of the electronic noises
- **Peak estimations** : followed by the low pass filter we used Savitzky-Golay smoothing filter (polynomial order 3, frame length 11) → this frame length would help us in much broader width analysis ( previous 5 data points + current point + next 5 points in the sliding window for the polynomial fit) this would help us in preserving the essential peaks of physical actions like footsteps





# Classification of Thresholds

coming to the classification of data , through our test analysis we have figured out thresholds for the classification of activities

## Sitting

Low acceleration ( $<10$ ) & a low orientation Y-axis tilt ( $<15$  degree indication of flat phone)

## Standing

Low acceleration ( $<10$ ) & high orientation Y-axis tilt ( $>45$  degree indication of vertical phone)

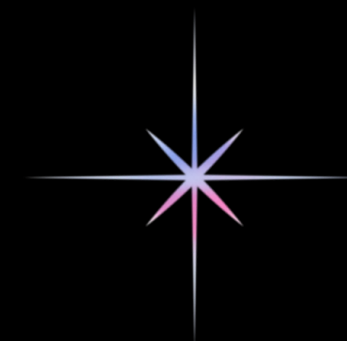
## walking

Medium acceleration (btw 10-11)

## Running

High acceleration ( $>11$ )





## Activity Scoring

### CAS(current activity score)

CAS evaluates real-time physical exertion using a weighted sum: 40% Mobility + 40% Intensity + 20% Direction Change. This provides an instant score of the user's current activity level

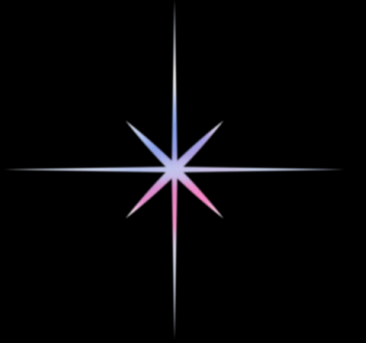
### AAS( average activity score)

AAS calculates the average of all current activity scores from the beginning of the timestamp up to the present moment. This provides a long-term estimate of the person's overall active period and exertion levels.

**Both the scores are updated and reported at the end of each second.**







## Saturation of parameters

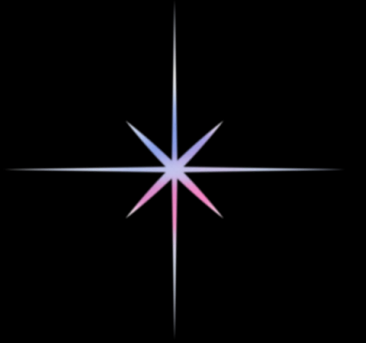
To prevent outliers & the raw spikes , we have saturated the values , this would help us in avoiding the skewing of final results

Saturation :  $x/x+k$

- Normalized Mobility:  $\text{mob\_n} = \text{Mobility} / (\text{Mobility} + 1)$  (Here,  $k=1$ ).
- Normalized Intensity:  $\text{int\_n} = \text{Intensity} / (\text{Intensity} + 50)$  (Here,  $k=50$ ).
- Normalized Direction:  $\text{dir\_n} = \text{Direction} / (\text{Direction} + 100)$  (Here,  $k=100$ ) .

Since from the observed data points we have figured out that the values of the parameters revolve around the used K values above , so we've adapted the Saturation functions through that.



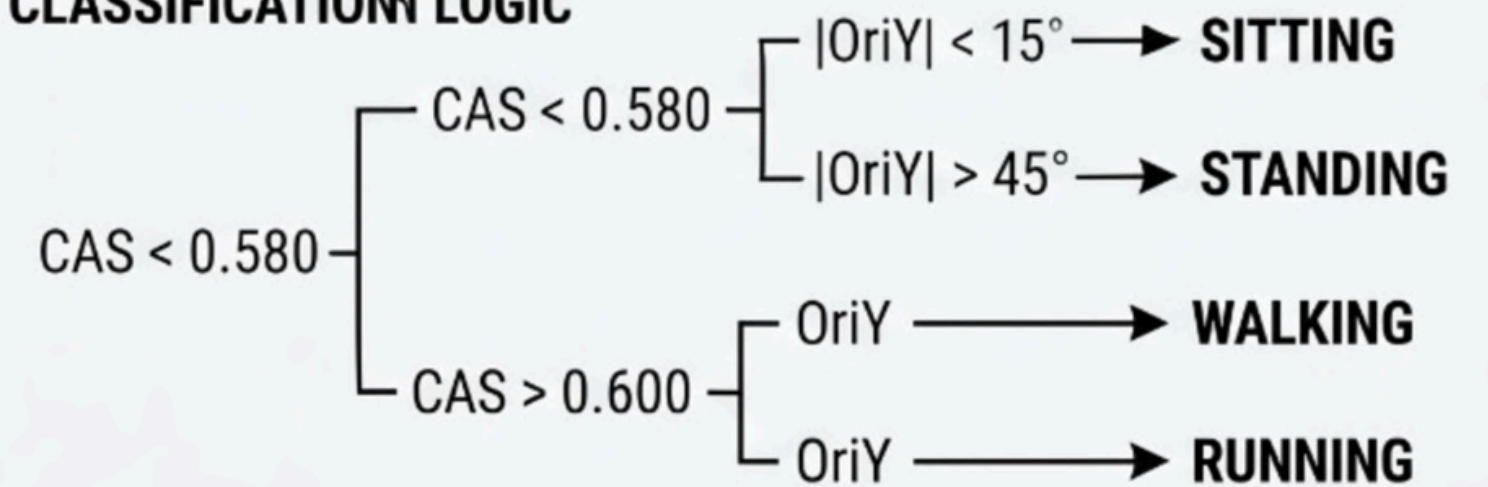


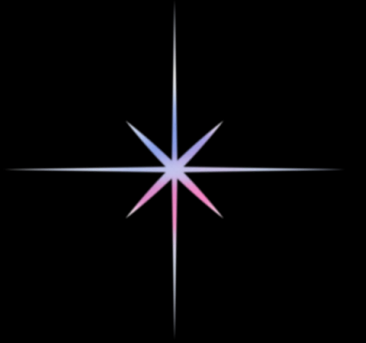
## Range of CAS

### Current Activity score range split

- Sitting: CAS  $< 0.60$  combined with a low orientation Y-axis tilt of  $< 15$  degrees (indicating the phone is flat)
- Standing: CAS  $< 0.60$  combined with a high orientation Y-axis tilt of  $> 45$  degrees (indicating the phone is vertical)
- Walking: CAS from 0.60 to 0.78 (Orientation is not factored in here, as the motion score itself is high enough to distinguish the activity)
- Running: CAS  $> 0.78$  (Orientation is not factored in)

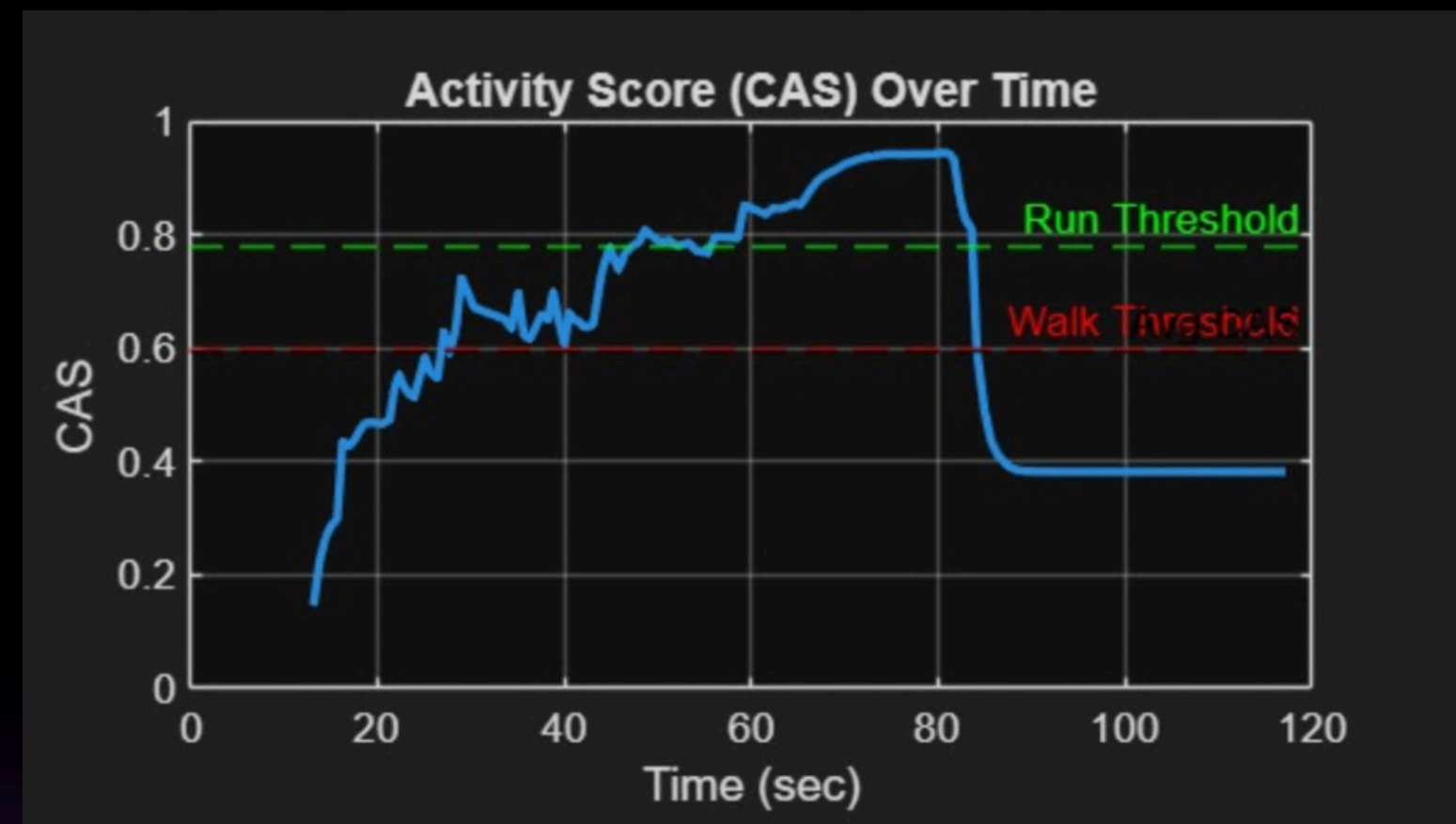
#### CLASSIFICATION LOGIC





# Why orientation not much in picture ?

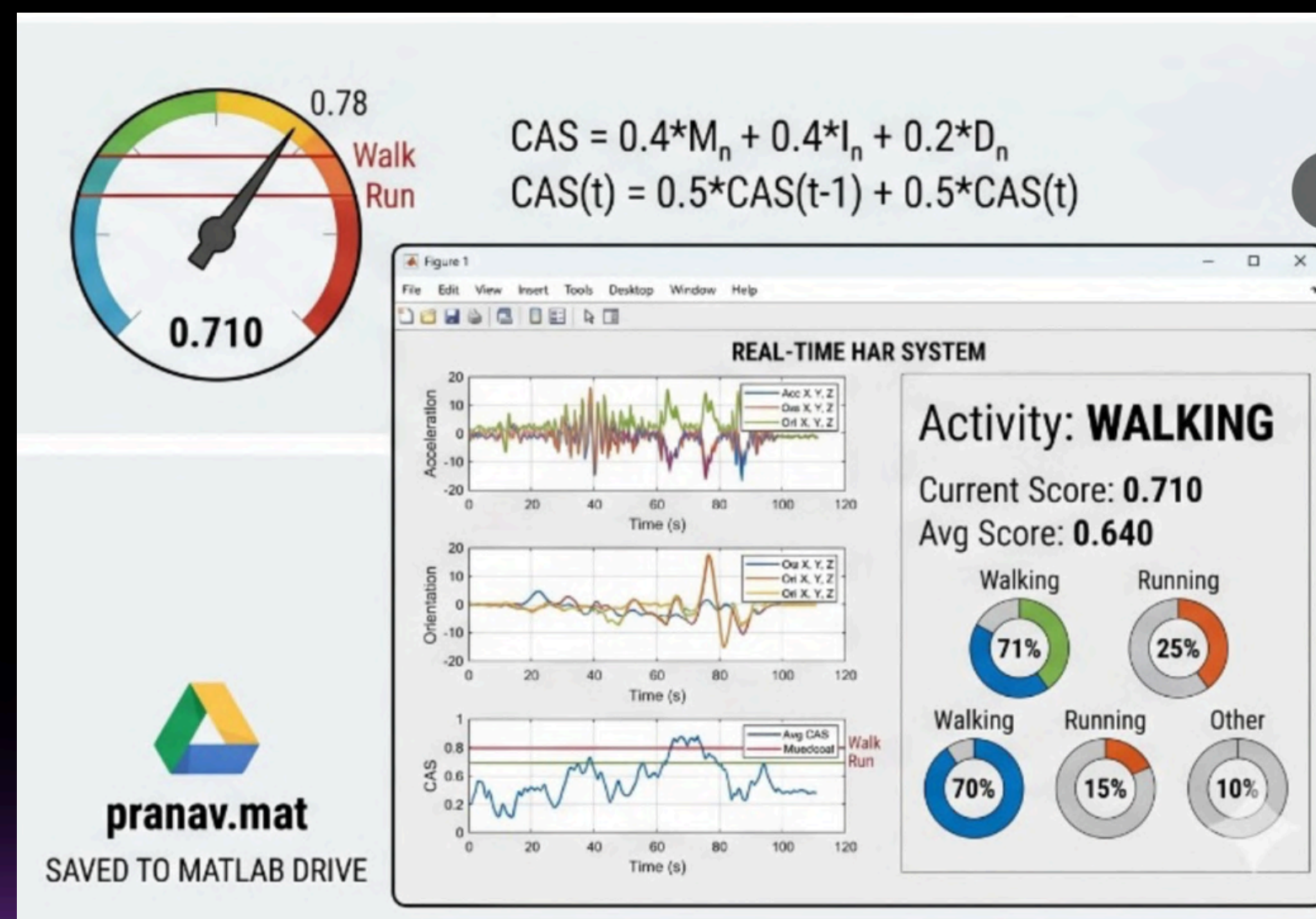
While calculating the CAS we have already given weightage to orientation and because of which while the orientation being used in the sitting and standing parts because the CAS difference isn't much , where as the running and walking are purely differentiated by the CAS scores



# Smoothening Current Activity score

$$CAS = 0.5 * CAS(t-1) + 0.5 * CAS$$

Here we've taken the average of CAS to you prevent the real-time graph from jumping erratically from one half-second to the next

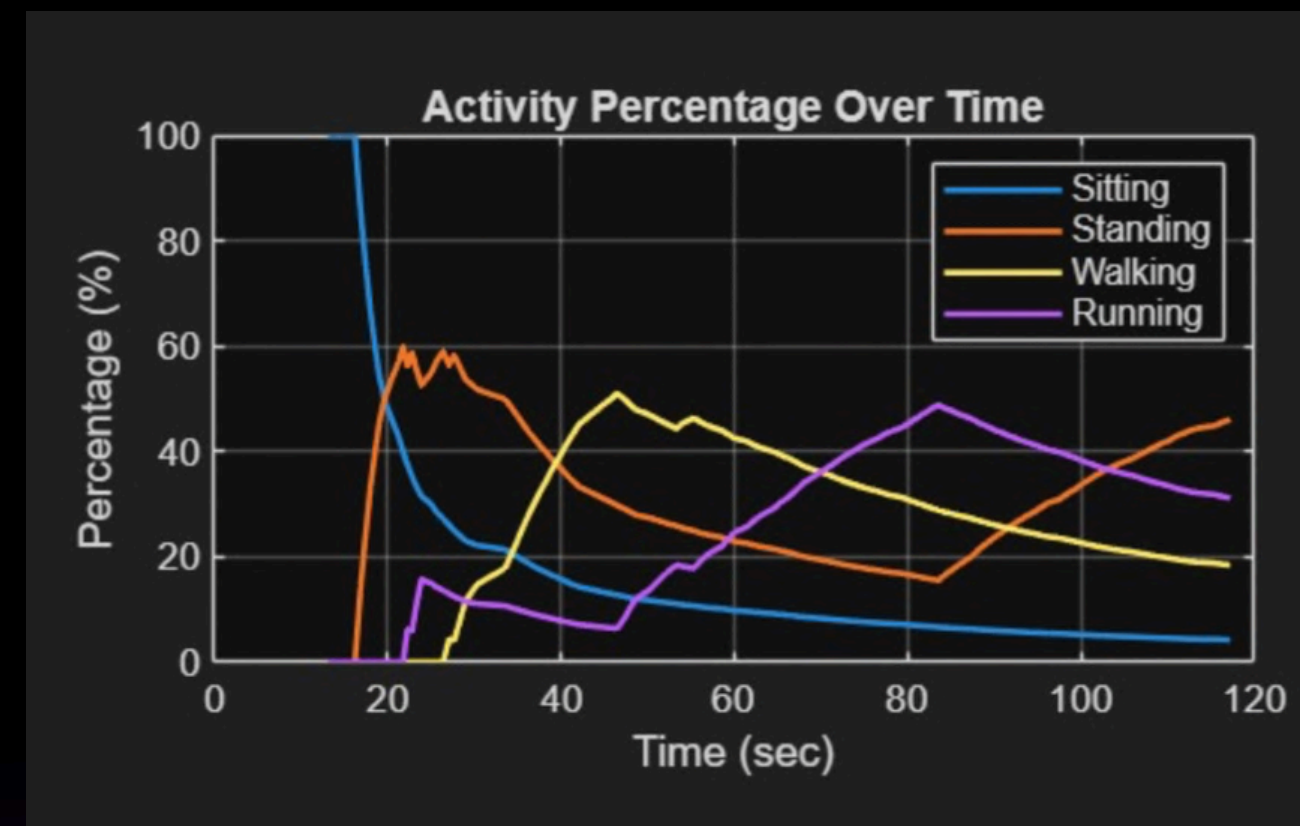


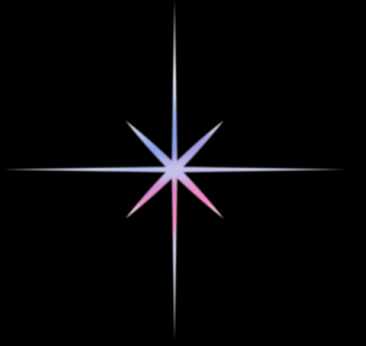


# Activity percentage tracking

Through our testing process the system monitors the duration of time spent in each activity

Cumulative Tracking: By dividing the time spent in each individual activity by the total elapsed time, the system generates a percentage breakdown of the entire session (e.g., Running: 32.9%, Walking: 27.8%, Standing: 22.2%, Sitting: 17.1%)



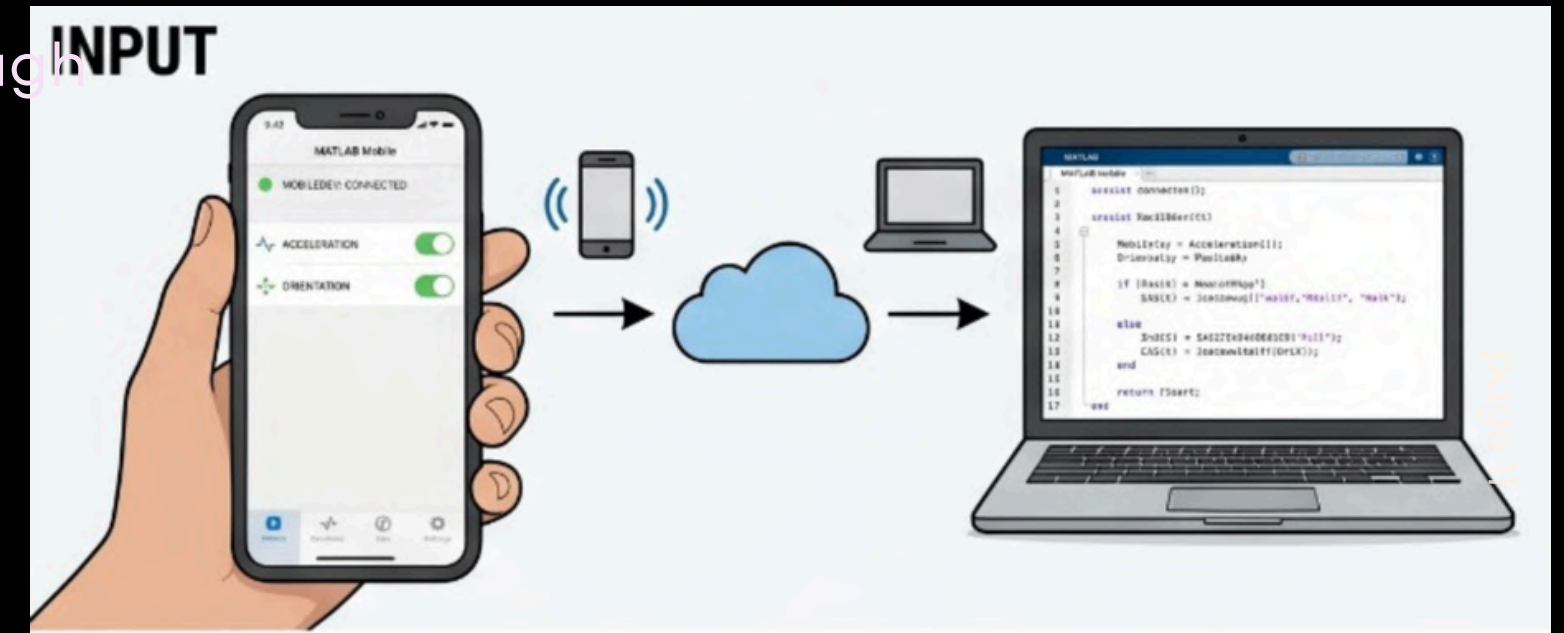


# Real-Time Edge Analytics & Use Cases

Through the data, our projects helps the users to receive immediate live feedback about their active states directly through the MATLAB GUI

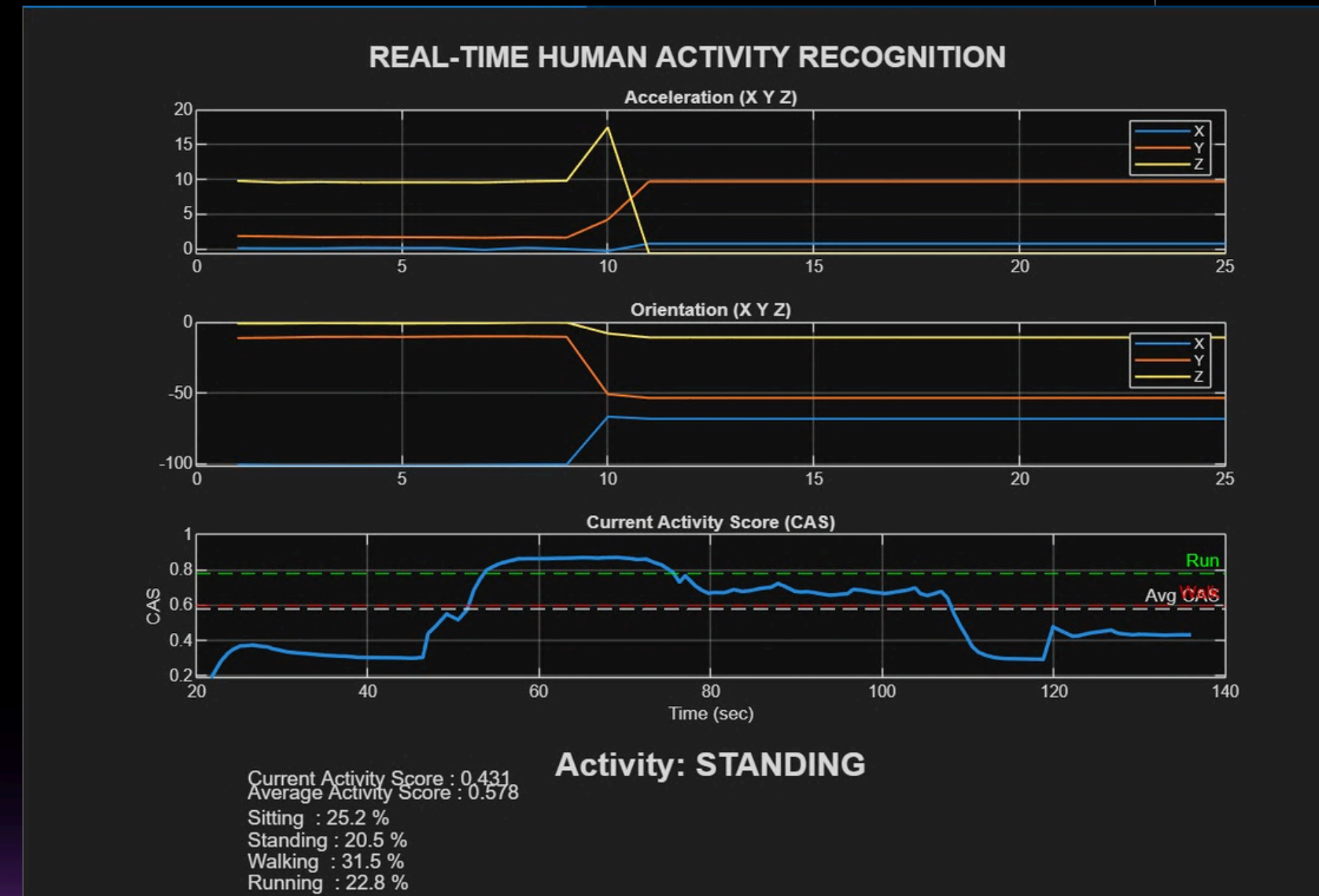
**Continuous Cloud Publishing:** The Processed results, including CAS values and activity percentages, are published and will be automatically saved as a .mat file to a HAR\_Project folder in the MATLAB Drive

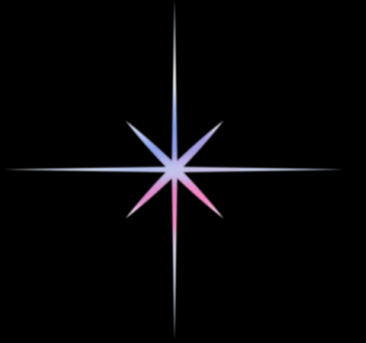
**Use Cases :** This "Edge-to-Cloud" pipeline enables long-term fitness tracking. Users can review historical data to monitor fitness progress over weeks or months, to access the user's data for deeper health insights .



# Over-View of Output

- **Sensor Waveforms:** The panels display the raw, real-time signal waveforms for both Acceleration (X, Y, Z) and Orientation (X, Y, Z).
- **CAS Trend Graph:** The third panel plots the Cumulative Activity Score (CAS) as a continuous line graph over time. It includes visual threshold lines (e.g., a green dashed line at 0.78 for running) and a dynamic line tracking the Average CAS.
- **Live Metrics Panel:** The bottom panel outputs the final computed intelligence in a human-readable text format. It distinctly highlights the current detected activity (e.g., "Activity: RUNNING"), displays the exact numerical values for the Current and Average Activity Scores, and provides a continuous percentage breakdown of the session (e.g., Running: 32.9%, Walking: 27.8%, Standing: 22.2%, Sitting: 17.1%)





# DrawBacks

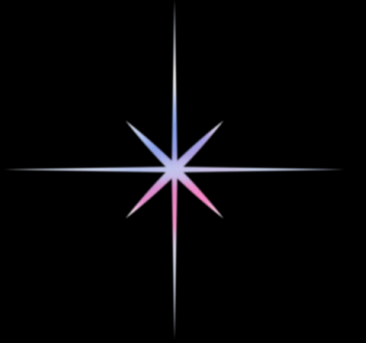
**Lower accuracy compared to ML-based models**

**Limited to basic activities (sit, stand, walk, run)**

**Hardcoded thresholds (not adaptive)**

**No personalization for different users**





# Advantages

**Real-time activity detection using mobile sensors**

**Lightweight system (no heavy ML required)**

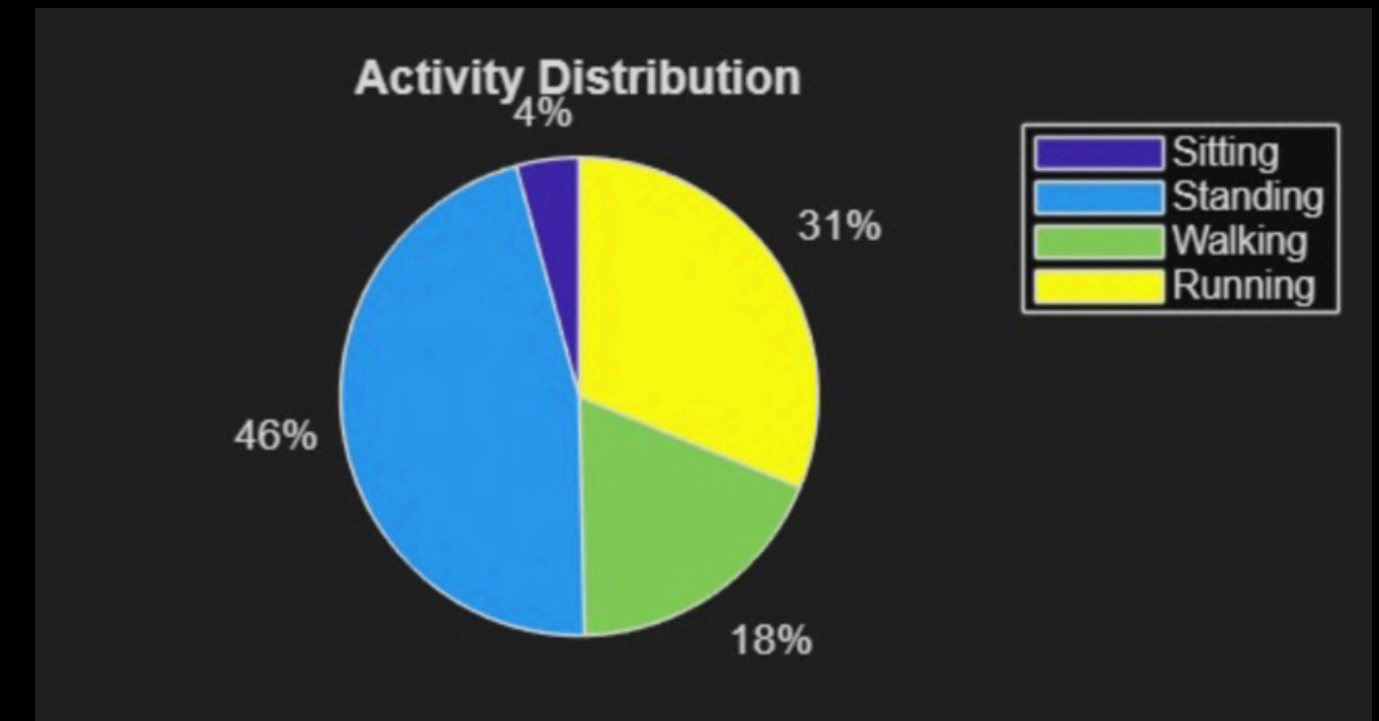
**Low latency and fast response**

**Easy to implement and understand**

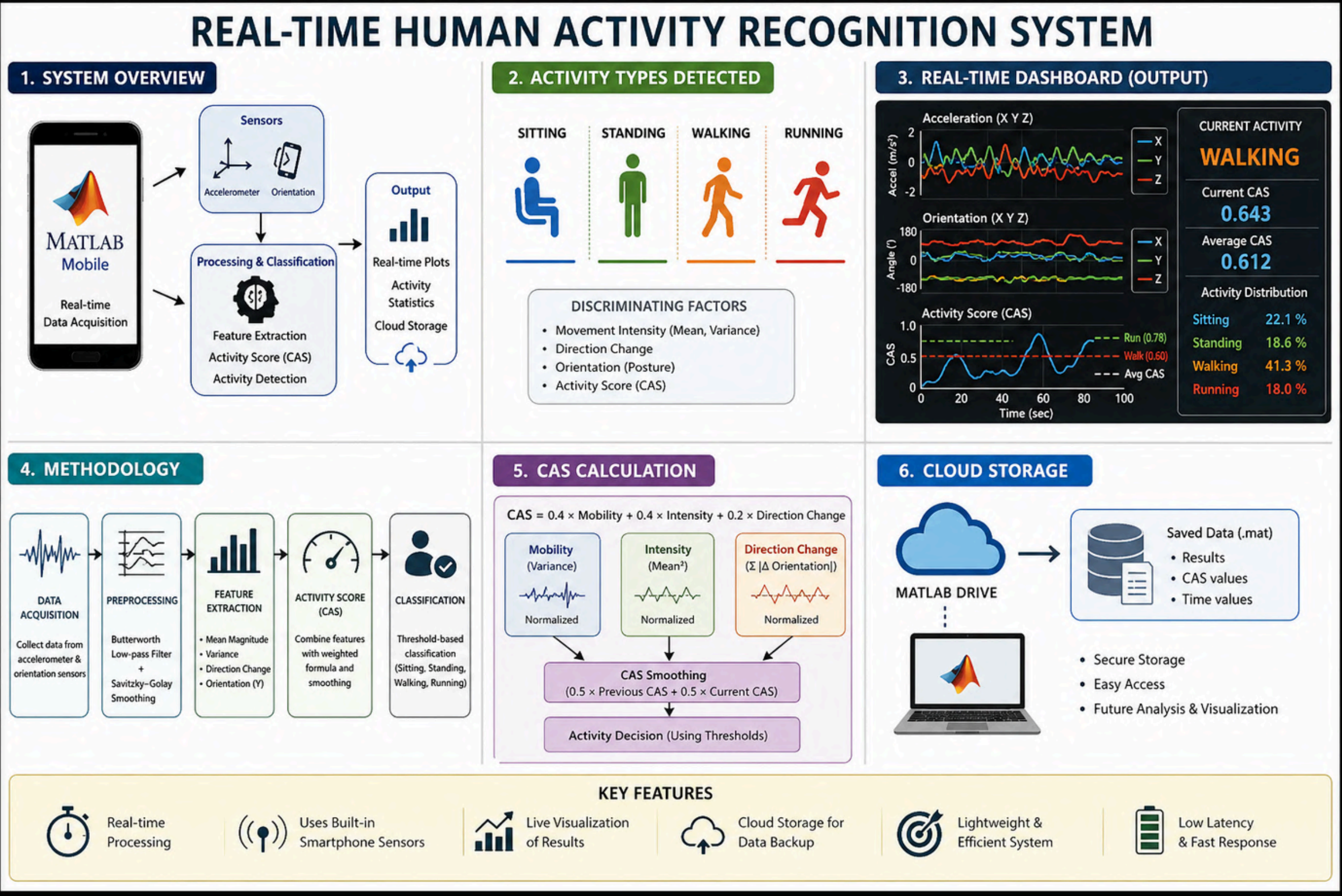
**Portable (works with smartphone only)**

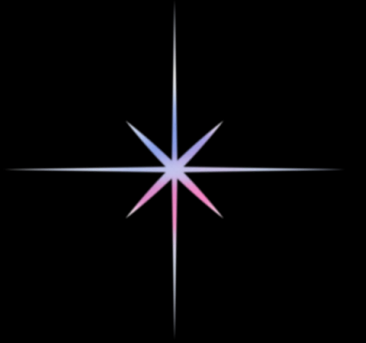
**Live visualization of sensor data and results**

**Cloud storage for data backup and analysis**



# Summary





# Future Scope

**Fall detection for safety monitoring**

**Step counter using acceleration peaks. ( for the steps calculation )**

**Calorie estimation based on activity intensity**

**Sedentary alerts (long sitting warning)**

**Machine Learning-based classification**



# THANK YOU!

